Name _____ Class _____ Date _____

# Task 1: Case study summaries

Read both summaries carefully. Use a highlighter to mark: (a) materials used, (b) how fire spread, (c) failures and cost-cutting.

# Grenfell Tower

London · 14 June 2017

## The building

A 24-storey social housing tower in North Kensington, London, built in 1974. In 2012-16, ACM (Aluminium Composite Material) cladding panels with a polyethylene (PE) core were fitted to the exterior. Mainly polyisocyanurate (PIR) foam insulation, with some phenolic insulation, was installed behind the cladding, with an air cavity between them.

## The fire

A fire involving a faulty fridge-freezer in a 4th-floor flat was reported at 00:54. Flames escaped through a window and ignited the ACM cladding. The combustible PE core was the main fuel for rapid external spread; the cavity and insulation formed part of the external wall system. The fire reached the roof in about 30 minutes. 72 people were killed.

## Key failures

- PE-core cladding was highly flammable. A proposed £293,368 saving concerned aluminium in place of zinc cladding, rather than a comparison of fire ratings.
- Manufacturers falsified or manipulated safety test data.
- Building control approved non-compliant cladding.
- Compartmentation failed; fire doors lacked effective self-closing devices.
- No sprinkler system; a single staircase; no effective evacuation plan.
- “Stay put” advice was maintained too long.
- Residents warned of more than 40 safety issues in 2016; their concerns were ignored.
- The Inquiry found decades of failure by government and industry.

Author: Wonyong Park (University of Southampton). Reviewers: Susan Crawford and Laura Hanley.

Sources and factual updates: teacher guide, pages 7 and 9.

Name _____ Class _____ Date _____

# Task 1: Case study summaries

Continue reading and highlighting the three categories from page 1.

# Wang Fuk Court

Tai Po, Hong Kong · 26 November 2025

## The building

Eight 31-storey residential towers in Tai Po, Hong Kong, built in 1983. About 4,600 residents, nearly 40% aged 65 or over. From 2024, all towers were undergoing facade renovation: bamboo scaffolding, green protective netting, and expanded polystyrene (EPS) foam boards over windows.

## The fire

The fire was reported at 14:51 on 26 November 2025. Reports described burning scaffolding netting and EPS foam boards, damaged windows allowing fire into apartments, and a continuous external fuel path allowing rapid upward spread. Wind-blown embers were reported to have crossed the roughly 7.5 m gap between towers. Seven of eight towers were engulfed. 168 people were killed, including one firefighter. Firefighting continued for more than 43 hours.

## Reported failures and concerns

- 2,300 rolls of non-compliant netting were reportedly bought at HK$54 per roll, compared with HK$100 for compliant netting. The implied price difference is HK$105,800 (about HK$106,000), not an audited total saving.
- After typhoon damage, cheap replacement netting was reportedly installed; compliant netting was placed at lower levels where inspectors could sample it.
- EPS foam boards over windows provided highly flammable fuel.
- Fire alarms were reported not to function; investigators examined suspected deactivation.
- Residents reportedly demonstrated foam flammability in September 2024; their warnings were dismissed.
- Sixteen inspections over sixteen months did not prevent the disaster.
- Safety certificates for netting were alleged to have been falsified.

## Investigation and legal status

By 10 June 2026: 35 arrests; seven individuals and two companies charged with 25 offences. On 9 September 2026, the Buildings Department announced 259 additional prosecutions. These are dated legal actions, not findings of guilt.

168 deaths confirmed by Police on 15 January 2026. Other reported details are distinguished from final findings. Sources: teacher guide, pages 7 and 9.

Name _____ Class _____ Date _____

# Task 2: Comparison table

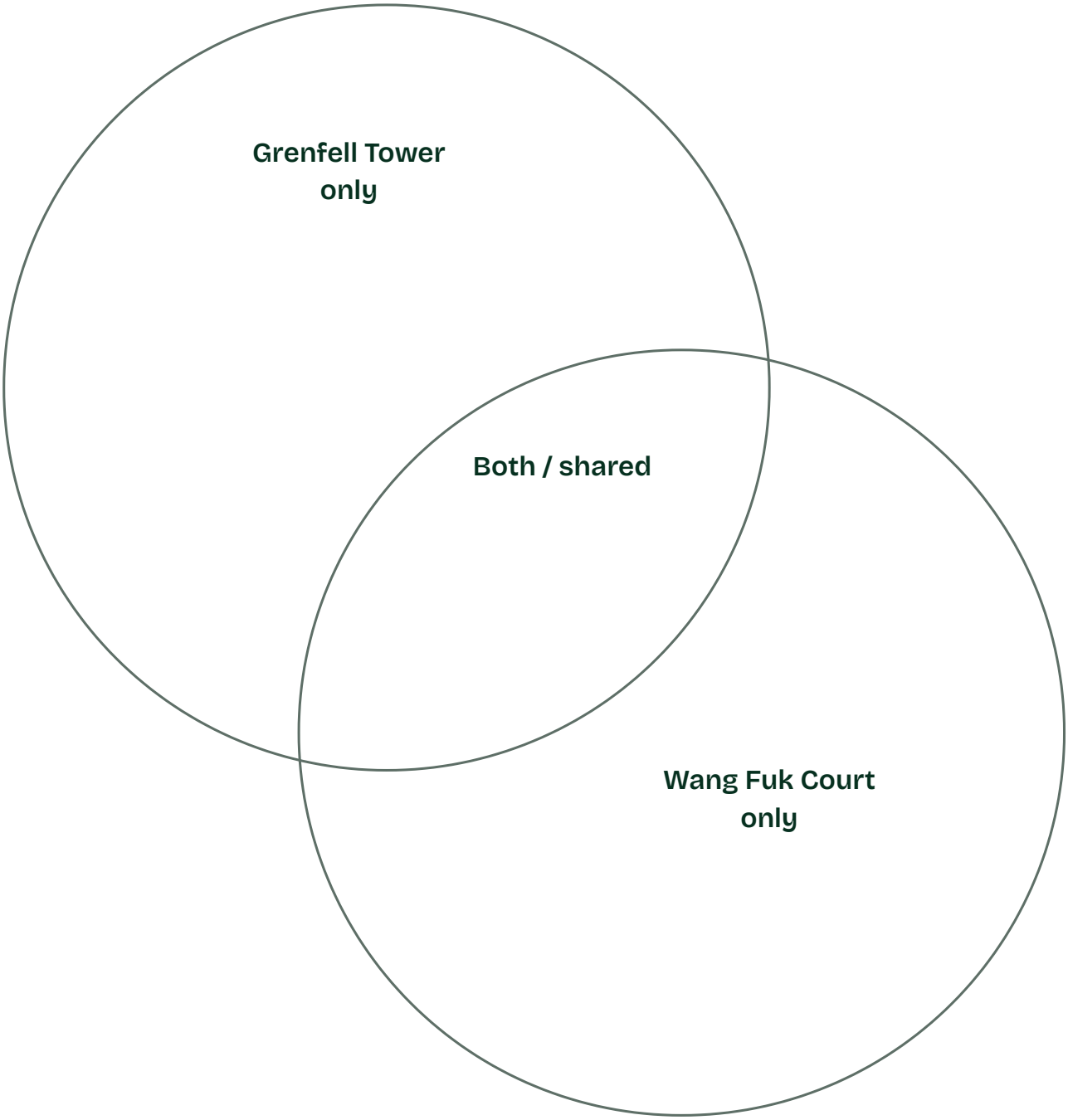
Complete the table below using the case studies in Task 1 and the information from the presentation slides.

| Factor | Grenfell Tower | Wang Fuk Court |
|---|---|---|
| What flammable material was applied to the exterior? | | |
| How did the fire spread upward so quickly? | | |
| How did the fire enter the apartments? | | |
| What cost-cutting decisions were made? | | |
| What warnings were ignored before the fire? | | |
| How did regulations or inspections fail? | | |
| What evacuation problems occurred? | | |

Name _____ Class _____ Date _____

# Task 3: Similarities and differences

Complete the Venn diagram below. Write factors unique to each fire in the outer sections, and shared factors in the overlapping centre. Aim for at least 3 points in each section.



Name _____ Class _____ Date _____

# Task 4: Evaluative writing

**Key question:** What do the Grenfell Tower and Wang Fuk Court fires tell us about the ethical responsibilities of engineers, designers, manufacturers, and regulators when making decisions about building materials and safety?

Write **6-8 sentences** answering the key question above. Use evidence from **BOTH** case studies. Try to use the sentence starters and connectives below to structure your response.

## Sentence starters and connectives:

- Both the Grenfell Tower and Wang Fuk Court fires demonstrate that...
- A key similarity between the two cases is...
- In the case of Grenfell Tower... Similarly, at Wang Fuk Court...
- This shows that engineers have a duty of care to...
- One ethical issue highlighted by both fires is...
- If manufacturers had not... then...
- The most significant lesson for designers and engineers is...
- In my own design practice, I should...

Your response